1. Define Intelligence

Intelligence is the ability to perceive, learn, reason, solve problems, make decisions, and act appropriately. In the notes, intelligence is associated with doing the right things and with skills such as perception, learning, linguistic intelligence, problem solving, and reasoning.

2. Define Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science that creates intelligent machines which can **behave like humans, think like humans, and make decisions** without being explicitly pre-programmed for every situation.

3. Name one application of AI and explain its significance

Application: Robotic vehicles / self-driving cars

AI is used in robotic vehicles to sense the environment using cameras, radar and laser rangefinders and to control steering, braking and acceleration. Its significance is that it enables vehicles to drive autonomously, follow traffic rules, and avoid pedestrians and other vehicles.

4. Which assessment is used to test the machine?

The **Turing Test** is used to test a machine's intelligent behaviour.

It was proposed by **Alan Turing in 1950**. A machine passes the test if a human interrogator cannot differentiate its responses from those of a human.

5. Differentiate Rationality and Omniscience

Rationality	**Omniscience**
A rational agent selects the best possible action based on the available percepts and knowledge.	An omniscient agent knows the actual outcome of its actions.
It does **not** require complete knowledge.	It requires knowledge of the actual outcome.
Rationality is possible in reality.	Omniscience is impossible in reality.
Example: making the best decision based on what the agent has observed.	Knowing exactly what will happen after an action.

The notes specifically state that a rational agent does not have to know everything; it should make the best decision based on what it has perceived.

6. Define Agent Function

An **agent function** maps any given **percept sequence to an action**. Mathematically, it is represented as:

$f: P \rightarrow A^*$

where **P*** represents percept sequences and **A** represents actions.

7. What are the operations in Vacuum Cleaner World?

The vacuum-cleaner world has two locations, **A and B**.

The possible operations/actions are:

1. **Move Left**
2. **Move Right**
3. **Suck up dirt**
4. **NoOp (No Operation)**

The agent perceives its location and whether the location is dirty, and then selects an appropriate action.

8. Why is chess game semi-dynamic?

Chess played **with a clock** is considered **semi-dynamic** because the environment (chessboard) remains unchanged while the agent is thinking, but the **agent's performance changes with time** because of the clock.

9. Expand and Define PEAS

PEAS stands for:

- **P – Performance Measure:** Objective used to measure the success of the agent.
- **E – Environment:** The surroundings in which the agent operates.
- **A – Actuators:** Devices used by the agent to perform actions.
- **S – Sensors:** Devices used by the agent to perceive the environment.

PEAS is a model used to describe the properties of an AI/rational agent.

10. How to measure the performance of a medical diagnosis system?

The uploaded notes **do not give a specific performance measure for a medical diagnosis system**.

The notes explain generally that a performance measure defines the success criterion of an agent and that the performance measure should be designed according to what is actually desired from the environment.

11. Define Agent Program

An **agent program** is the implementation of the agent function. It executes on the physical architecture of the agent and produces the required actions based on percepts.

12. Define Problem

A **problem** in AI is defined using the components required by a problem-solving agent:

1. Initial State
2. Actions
3. Transition Model
4. Goal Test
5. Path Cost

Together, these components define the problem that is given to a problem-solving algorithm.

13. Define Abstraction

Abstraction is the process of removing irrelevant details from the real-world problem and representing only the important details in the state space and actions.

An abstraction is **valid** if it can be expanded into a solution in the more detailed world, and it is **useful** if carrying out the solution is easier than solving the original problem directly.

14. Describe the concept of Rationality in AI

Rationality means selecting the best possible action to achieve the desired performance.

A rational agent selects an action that is expected to **maximize its performance measure**, based on:

1. The performance measure.
2. The agent's prior knowledge.
3. The actions available to the agent.
4. The percept sequence received so far.

A rational agent does not need to know everything. It makes the best decision based on the information available to it.

15. Define Learning Agent

A **learning agent** is an AI agent that can **learn from its past experiences** and improve its performance.

A learning agent starts with basic knowledge and can adapt automatically through learning. It has four main components:

1. **Learning Element** – improves performance by learning from the environment.
2. **Performance Element** – selects actions based on percepts.

3. **Critic** – provides feedback about how well the agent is performing.
4. **Problem Generator** – suggests actions that provide new and useful experiences.

16. Write two components of a problem-solving agent

Two important components/steps of a problem-solving agent are:

1. **Goal Formulation** – determines the goal that the agent wants to achieve.
2. **Problem Formulation** – determines the states and actions that should be considered to achieve the goal.

The overall problem-solving process is **formulate → search → execute**.

17. List the differences between strong AI and weak AI?

Strong AI	**Weak AI**
Strong AI aims to create machines with **human-like intelligence and cognitive abilities**.	Weak AI is designed to perform **specific tasks** intelligently.
It aims to **think, reason, learn, and understand** like a human.	It generally performs tasks based on its **specific programming or training**.
It would have **general-purpose intelligence**.	It has **limited/specific-purpose intelligence**.
Strong AI is still **theoretical/hypothetical**.	Weak AI is **widely used in real-world applications**.
Example: a hypothetical machine with human-level general intelligence.	Examples include **chatbots, recommendation systems, and voice assistants**.

In short:

Strong AI = General human-like intelligence

Weak AI = Intelligence designed for specific tasks